

DRY-RUN GUIDE | gym equipments.py  
Source: ca-2\psup\gym equipments.py

#### CORE BEHAVIOR

Stores equipment counts in a dictionary, applies a booking increment, adds/removes entries safely, queries usage, then prints total and busiest remaining equipment.

#### REPRESENTATIVE INPUT

Initial treadmill=10,bike=5,rower=8; book treadmill +2; add mats=3; remove bike; query bike.

#### STEP TRACE

treadmill=12; add mats=3; pop bike=5; lookup removed bike defaults to 0; remaining total=12+8+3=23; treadmill is busiest.

#### EXPECTED OUTPUT / RETURN

Progress messages; final 'Total users: 23' and 'Busiest equipment: treadmill (12)'.

#### ACTUAL SOURCE EXCERPT

Verbatim excerpt from ca-2\psup\gym equipments.py; line numbers refer to the original file. No source was modified.

```
001 | # Gym Equipment Tracker
002 | # Uses a dictionary as the only data type and avoids loops.
003 |
004 | equipment = {}
005 |
006 | # Read 3 pieces of equipment directly into the dictionary
007 | name1 = input().strip()
008 | users1 = int(input().strip())
009 | equipment[name1] = users1
010 | print(f"{name1}: {users1} users")
011 |
012 | name2 = input().strip()
013 | users2 = int(input().strip())
014 | equipment[name2] = users2
015 | print(f"{name2}: {users2} users")
016 |
017 | name3 = input().strip()
018 | users3 = int(input().strip())
019 | equipment[name3] = users3
020 | print(f"{name3}: {users3} users")
021 |
022 | # Booking update using dictionary-safe increment
023 | booking_name = input().strip()
024 | amount = int(input().strip())
025 | equipment[booking_name] = equipment.get(booking_name, 0) + amount
026 | print(f"Updated {booking_name}: {equipment[booking_name]} users")
027 |
028 | # Add new equipment to the dictionary
029 | new_name = input().strip()
030 | new_users = int(input().strip())
031 | equipment[new_name] = new_users
032 | print(f"Added {new_name}: {new_users} users")
033 |
034 | # Remove equipment from the dictionary safely
```

```
035 | removed_name = input().strip()
036 | removed_users = equipment.pop(removed_name, 0)
037 | print(f"Removed {removed_name} with {removed_users} users")
038 |
039 | # Query equipment usage without raising KeyError
040 | query_name = input().strip()
041 | usage_value = equipment.get(query_name, 0)
042 | print(f"Usage of {query_name}: {usage_value} users")
... 13 later source lines omitted; inspect the original.
```

#### DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.